

Distributed Electric Propulsion: A Novel Method for Noise Reduction and Thermal Management

Aaryan Lath, Hayden Gunckel and Mia Drago
Purdue University, West Lafayette, IN, 47906

This paper surveys the critical technologies required for Distributed Electric Propulsion in Urban Air Mobility (UAM), focusing on the coupled challenges of acoustic signature reduction and thermal management. While DEP enables noise mitigation through low tip-speed rotors and aero-propulsive coupling, the high-power requirements of vertical flight introduce severe thermal loads that constrain vehicle performance. We review recent experimental and computational literature regarding propeller-wing interaction noise, wake impingement in tandem configurations, and advanced Battery Thermal Management Systems. The analysis highlights a fundamental design trade-off where the weight of effective liquid or hybrid cooling systems increases disk loading, potentially negating acoustic benefits. Future multi-disciplinary optimization must simultaneously address aerodynamic interference and transient thermal limits to achieve community acceptance.

Nomenclature

A rotor disk area, m^2	r observer distance, m
A_R receiver absorption area, m^2	r_m reference distance, m
A_r distance attenuation coefficient, dB	S surface area, m^2
A_α atmospheric absorption coefficient, dB	SEL sound exposure level, dB
a speed of sound, m/s	SOC state of charge, %
B number of blades	S_{pp}^{TE} trailing edge power spectral density, Pa^2/Hz
c blade chord length, m	T temperature, K or $^\circ C$
C_T thrust coefficient	T_{thrust} thrust force, N
F Faraday's constant, C/mol	t_e effective time, s
I current, A	V volume, m^3
I_{noise} acoustic intensity, W/m^2	W weight, N or kg
L_i indoor sound pressure level, dB	α atmospheric absorption coefficient, dB/m
L_m measured sound level at reference distance, dB	ΔR radial segment length, m
L_o outdoor sound pressure level, dB	ΔS entropy change, J/(mol·K)
L_p sound pressure level, dB	θ directivity angle, deg
L_w sound power level, dB	ϕ azimuthal angle, deg
l_y spanwise correlation length, m	Ω rotor angular velocity, rad/s
M_{tip} blade tip Mach number	ω angular frequency, rad/s
m_{BTMS} battery thermal management system mass, kg	
n number of electrons transferred	
p acoustic pressure fluctuation, Pa	
p_B broadband noise pressure, Pa	
p_L loading noise pressure, Pa	
p_N narrowband noise pressure, Pa	
p_S separation noise pressure, Pa	
p_T thickness noise pressure, Pa	
$\dot{Q}_{gen}$ heat generation rate, W	
R sound reduction index, dB	
R_{int} battery internal resistance, Ω	

Abbreviations

BPF	=	blade passage frequency
BTMS	=	battery thermal management system
CFD	=	computational fluid dynamics
DEP	=	distributed electric propulsion
eVTOL	=	electric vertical take-off and landing
FW-H	=	Ffowcs Williams-Hawkings
GA	=	genetic algorithm
LiB	=	lithium-ion battery
LSAWT	=	Low Speed Aeroacoustic Wind Tunnel
MDO	=	multi-disciplinary design optimization
MTOW	=	maximum take-off weight
NFAC	=	National Full-Scale Aerodynamics Complex
OASPL	=	overall sound pressure level
PAS	=	Propeller Analytic System
PCM	=	phase change material
PMC	=	polymer matrix composite
SPL	=	sound pressure level
SPSA	=	Simultaneous Perturbation Stochastic Approximation
sUAS	=	small unmanned aerial system
TEC	=	thermoelectric cooler
UAM	=	urban air mobility
UAV	=	unmanned aerial vehicle
URANS	=	unsteady Reynolds-averaged Navier-Stokes
VL	=	vertical landing
VTO	=	vertical take-off

I. Introduction

THE integration of Urban Air Mobility (UAM) into civil transportation infrastructure depends on achieving community acoustic acceptance. Unlike conventional aviation, Electric Vertical Take-Off and Landing (eVTOL) aircraft operate in close proximity to densely populated environments, making acoustic signature a primary design constraint. Distributed Electric Propulsion (DEP) has emerged as the primary architectural solution. By decoupling power generation from thrust generation, DEP allows for the distribution of thrust across multiple smaller propulsion systems.

In unmanned aerial vehicles and UAM platforms, electric propulsion systems (EPS) have proven to be more efficient, precise regarding control, reliable, and quieter than typical internal combustion engines [1]. Electric systems transform stored electric power into mechanical power using batteries to power motors that turn propellers. While batteries currently provide lower specific energy and have a slower dynamic response compared to fuel, the use of super capacitors and hybrid architectures helps mitigate this gap. However, the transition to high-voltage DEP introduces new design challenges. The high-power discharge rates required for vertical flight generate significant heat within the battery packs and electric motors [2].

Recent experimental testing of the Joby Aviation S4 propeller at the National Full-Scale Aerodynamics Complex (NFAC) demonstrated that noise generation is non-linear and highly sensitive to operational parameters. Schatzman et al. found that reducing RPM does not always reduce noise; specific “acoustic sweet spots” exist where blade pitch and RPM minimize noise for a given thrust [3]. This finding underscores the necessity for propulsion systems that can dynamically adapt to these non-linear acoustic optima. However, operating at these optimal points is often constrained by the thermal limits of the electric motors and batteries. Thus, the design of the Battery Thermal Management System (BTMS) becomes a critical acoustic variable. A heavy cooling system imposes a weight penalty that forces propulsors to operate at higher, louder thrust settings, while a lightweight system may limit high-power noise-abating maneuvers [4].

II. Literature Review

A. Modeling Acoustic Signatures

Using field or lab measurements we can determine the acoustics using sound propagation models. One variable Torija et al. focused on was the sound pressure level of an operating unmanned aerial vehicle during various maneuvers [5]. They divided the sound pressure levels into two categories: the sound heard a certain distance away outside and the sound heard indoors. They determined that acoustics as they propagate are a function of environmental conditions, distance, and flight maneuverability. Comprehensive measurement models consider environmental conditions, sound absorption coefficient, α , and distance as functions of the known receiver distance r . Measurements cover maneuvers such as hover, take-off, and landing which have been proven as the loudest operations due to their proximity to populated areas. Two assumptions were made to simplify the model: the drone would act as a single point source that propagates exponentially, the motor generators would be significant noise sources, and the atmospheric sound absorption coefficients A_α and A_r are the main modifying factors during the unmanned aerial vehicle acoustic propagation. The model for sound pressure level of an operating unmanned aerial vehicle is shown in Eq. 1.

$$L_p(r, \alpha) = L_w + A_r + A_\alpha \quad (1)$$

The sound power, L_w , at a reference distance, r_m , can be found by deriving the power from a certain reference point. The atmospheric sound absorption coefficients $A_{\alpha r_m}$ and A_{r_m} are the modifying factors during the unmanned aerial vehicle acoustic propagation. The equation is shown in Eq. 2.

$$L_w = L_m - A_{r_m} - A_{\alpha, r_m} \quad (2)$$

Using Eq. 2 we can approximate the pressure level in Eq. 3.

$$L_p(r) = L_m - 20 \log_{10} \left(\frac{r}{r_m} \right) - L_{atm} \quad (3)$$

More specifically the atmospheric absorption terms L_{atm} can be found exponentially by Eq. 4.

$$L_{atm} = 10 \log_{10} e^{2\alpha r} \quad (4)$$

Lastly, the sound exposure level, SEL , can be found if the maximum absorption term LA_{max} is known. Using effective time t_e , SEL is found using Eq. 5.

$$SEL = LA_{max} + 10 \log_{10} \left(\frac{t_e}{t_0} \right); \quad t_0 = 1 \text{ s.} \quad (5)$$

The team collected data over time and compared it to known results. It was compiled in Fig. 1. It was found that rotor speed and pitch made the most impact.

Next the team applied this model to indoor acoustics. First you must add building reduction effects. The sound reduction index R is found by taking the difference between sound levels at the source, the UAV, and the receiver. The absorption factor A_R depends on frequency content, building materials, window type, and condition.

$$R = L_S - L_R + 10 \log_{10} \left(\frac{S}{A_R} \right) \quad (6)$$

The equation below is an updated propagation model in the receiver environment volume V and surface area S for time T that takes building facade into account.

$$L_i = L_o - R' + 10 \log_{10} \left(\frac{S}{0.16V} \right) \quad (7)$$

Overall the impact of drone acoustics for indoor receivers is a gap in current techniques for noise evaluations. Drones have special acoustic characteristics such as pure tone and high frequency broadband noise. These characteristics make it difficult to use known noise assessment methods used for traffic or other aircraft.

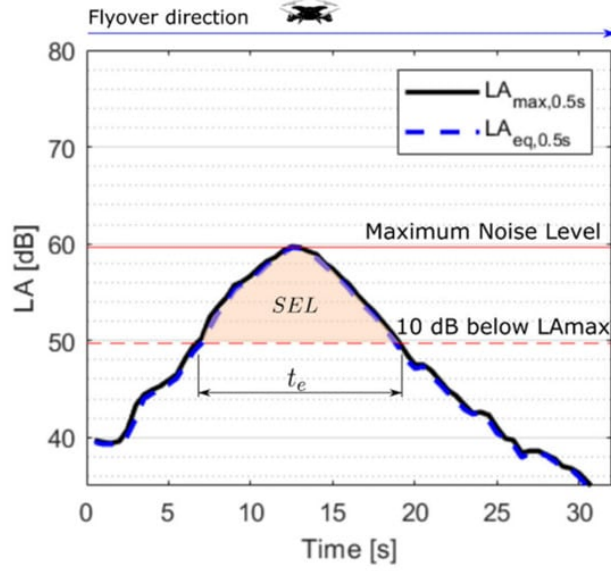


Fig. 1 Sound exposure level, maximum noise level, and effective time. The presented data corresponds to a recording during a fast flyover operation of the drone Typhoon.

B. Aerodynamic Integration and Acoustic Signatures

The distribution of propulsion units across the airframe alters the aerodynamic environment, creating interacting noise sources. In tandem tilt-wing configurations, the wake from upstream rotors significantly affects downstream components. Liu et al. utilized high-fidelity Computational Fluid Dynamics (CFD) to investigate these interactions during conversion flight [6]. They reported a 2.1% reduction in thrust for rear rotors due to wake impingement and a 27.7% reduction in rear wing lift due to secondary downwash. This aerodynamic degradation forces rear rotors to operate at higher power settings to maintain trim, directly increasing tonal noise and blade-vortex interaction (BVI) noise.

Further research by TU Delft on propeller-propeller interaction identified severe penalties in “one-after-another” configurations typical of cruise flight. In worst-case alignment, the rear propeller can suffer thrust losses up to 80%, requiring power increases of up to 30% to compensate [7]. To mitigate these penalties, concepts like the “Jetfoil” integrate high-solidity ducted fans into the wing structure. Ahuja et al. demonstrated that such integration provides clean inflow to the fans even at high angles of attack, minimizing turbulence ingestion noise [8].

The propeller aerodynamic noise is made up of two main components: narrow and broadband components. Pagliaroli et al. define the total pressure fluctuation as the combination of narrowband component pressure fluctuations $p_N(x, t)$ and broadband component pressure fluctuations $p_B(x, t)$ [9].

$$p(x, t) = p_N(x, t) + p_B(x, t) \quad (8)$$

The narrowband given by p_N noise arises from periodic blade motion. It is generated by the blade rotation at the frequency and magnitude of rotational speed. This noise is further broken down into two parts: noise due to thickness of the blade and noise due to the aerodynamic loading on the blade. Thickness noise, p_T , refers to the displacement of air by the blade’s physical shape. The loading noise, p_L , is due to the unsteady force distribution over the body. For thin blades and subsonic flight the relation of thickness noise and aerodynamic loading noise is given by Eq. 9.

$$p_N = p_T + p_L \quad (9)$$

Broadband noise arises from turbulent flow interactions. The noise is generated at the blade’s leading edge, trailing edge, or blade tip. It is composed of the noise of turbulence of incoming flow p_L , the noise from the turbulent boundary layer over the blade surface, p_T , and the noise caused by laminar separation of the flow, p_S .

$$p_B = p_T + p_L + p_S \quad (10)$$

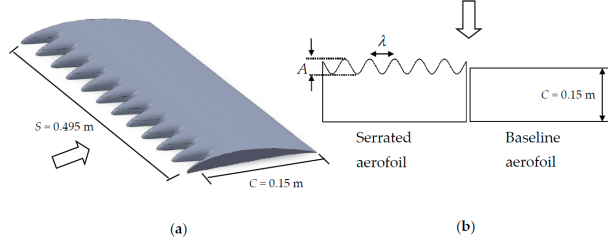


Fig. 2 The serrated trailing edges also known as feathered blades are used on wind turbine blades already to reduce noise.

The trailing edge broadband noise is coupled with Power Spectral Density and the noise caused by the turbulent boundary can be found in relation to the correlation length l_y . The power spectral density of the trailing noise S_{pp}^{TE} is a function of r , the observer position vector, ω , the angular frequency, B , the number of blades, c , the chord, a , the speed of sound, and the directive function.

$$S_{pp}^{TE}(r, \theta, \omega) = \frac{B}{8\pi} \left(\frac{\omega c}{2ar} \right)^2 \Delta RD(\theta, \phi) |I|^2 \Phi_{pp|y} \quad (11)$$

Pagliaroli et al. determined that the spanwise correlation length is key to controlling the trailing edge noise of a UAV [9]. One way researchers have altered the spanwise length is by including a saw tooth cut on trailing edges. By doing this in theory, the trailing edge has a smaller spanwise length. The serrations have critical constraints. A minimum tooth height requirement of 0.25 ensures that serrations interact with large turbulent entities in the boundary layer. The sharpness must be greater than 45 degrees, and the Strouhal number must be greater than one to make a difference in noise reduction. There are tradeoffs with this proposed design: the increase of serration height reduces trailing edge wake intensity but has aerodynamic penalties.

There are three tradeoffs to consider when optimizing geometry. Torija et al. describe current methods that maximize efficiency by minimizing power for thrust [5]. Acoustic performance was not considered. New design focuses on optimization that includes aerodynamics, acoustics, and structural integrity.

The Gur and Rosen model integrated acoustic cost functions with limits on power and stress [10]. The limits provide feasible chord and radius geometries.

Overall it was found that increasing chords can reduce tonal noise but increases vibration due to decrease in blade thickness. However, with an increase in vibration the propeller will need to be made from stronger materials as this increases the risk of structural failure. The pitch angle highly affects broadband noise. At higher pitch angles the flow is more likely to separate causing more separation bubbles to interact. To mitigate this effect, tip geometry is a quick way to produce quieter noise. Optimizing the tip will cause less turbulence resulting in less broadband noise as explained earlier in this review. Lastly, unequal blade spacing can reduce tonal noise without affecting thrust. As this will reduce the tonal noise it becomes a structural issue. The Gur and Rosen model takes these key design tradeoffs into account to determine the optimal propeller to reduce acoustic signatures.

Gur and Rosen tested different configurations of different propeller geometries and power capabilities for two blade propellers [10].

C. Battery Thermal Management Systems (BTMS)

The acoustic profile of eVTOL aircraft is fundamentally coupled to vehicle mass and the thermal management architecture. As DEP systems demand high power during vertical flight phases, managing the resultant heat generation in battery packs becomes critical. This subsection surveys the principal BTMS technologies reported in the literature, categorizing approaches by their underlying heat transfer mechanisms.

1. Passive Cooling Approaches

Passive BTMS architectures rely on natural convection, radiation, or latent heat storage without requiring external power input. Nelson and Almansour at NASA Glenn Research Center explored lightweight active air cooling using polymer matrix composite (PMC) packs with helical fins [11]. This design enhances convective heat transfer through geometric optimization while reducing weight by 0.04 kg per module compared to traditional aluminum designs,

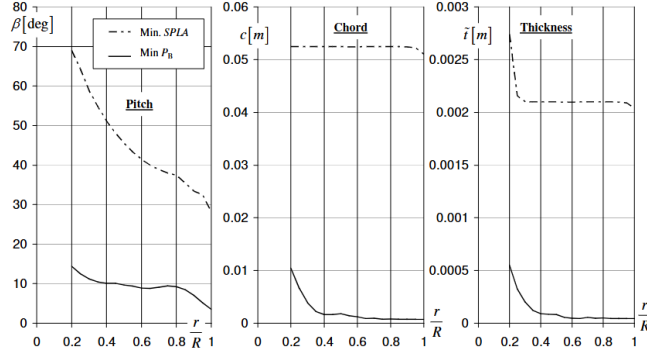


Fig. 3 Geometry of Optimal Propeller

increasing energy density by 8.25 Wh/kg. The helical fin geometry disrupts thermal boundary layer growth, improving heat transfer coefficients compared to straight-fin configurations.

Phase Change Materials (PCM) offer an alternative passive approach by absorbing heat through latent heat of fusion. Several studies have examined paraffin-based and hydrated salt PCMs for battery thermal management, with particular attention to the transient heat absorption capability that aligns well with the “peaky” load profiles of eVTOL missions [2, 12]. However, PCM-only systems face challenges including low thermal conductivity of the base material and the need for heat rejection mechanisms to re-solidify the PCM between flight cycles.

Heat pipes represent a third passive technology, utilizing two-phase heat transfer to achieve high effective thermal conductivity. Wang et al. investigated flat heat pipes combined with ram air cooling for eVTOL applications, demonstrating effective heat transport during cruise phases when sufficient airflow is available [13].

2. Active Cooling Approaches

Active BTMS architectures employ external power to achieve higher heat rejection rates. Liquid cooling systems using pumped coolant loops have been extensively studied for eVTOL applications. Xiong et al. developed wavy-channel liquid cooling technology demonstrating that liquid-based systems can maintain acceptable battery temperatures with system mass less than 20% of battery pack weight [14]. Chen et al. optimized liquid-cooled BTMS for tilt-wing configurations, examining effects of cruise altitude, hover duration, and battery oversizing on system weight and battery lifetime [15].

Thermoelectric Coolers (TECs) based on the Peltier effect offer precise temperature control without moving fluids. Kellermann et al. investigated TEC-based cooling for a 19-seat hybrid-electric aircraft, providing comprehensive analysis of the system-level weight implications [16]. Zhang et al. extended this work to eVTOL-specific hot-day operations, examining the tradeoffs between thermal capability and vehicle weight [17].

3. Hybrid Cooling Approaches

Recognizing limitations of single-technology approaches, researchers have developed hybrid architectures combining multiple mechanisms. Chai et al. validated a hybrid BTMS combining PCM with dielectric fluid immersion cooling [4]. This system reduced maximum cell temperatures by 32.4% compared to air cooling while maintaining PCM mass at only 5.32% of total battery pack weight. The hybrid approach exploits PCM for transient heat absorption during takeoff while using immersion cooling for steady-state heat rejection and PCM regeneration.

Dong et al. proposed integrating topology-optimized air-cooling fins with PCM, demonstrating reduced fan energy consumption under various control strategies [18]. These hybrid approaches explicitly target the transient nature of eVTOL thermal loads, designing systems to “shave” thermal peaks rather than sizing for maximum instantaneous heat rejection.

4. Optimization and Algorithmic Design

The multi-objective nature of BTMS design—balancing thermal performance, weight, power consumption, and cost—has motivated algorithmic optimization approaches. Wiston et al. employed hybrid Genetic Algorithm and Simultaneous Perturbation Stochastic Approximation (GA+SPSA) optimization to navigate the BTMS design space,

achieving 22% reduction in peak temperatures compared to baseline designs [19]. Guibert et al. applied thermo-mechanical topology optimization to battery pack structural design, simultaneously addressing thermal and mechanical constraints [20]. The implications of these various approaches for DEP vehicle design, particularly regarding the coupling between BTMS mass and acoustic signatures, are examined in Section III.

D. Intelligent Control and Optimization

Advanced control strategies are required to manage the coupling between thermal safety and acoustic targets. Shili et al. developed a “Novel Intelligent Thermal Feedback Framework” that uses machine learning to estimate motor winding temperatures in real-time [21]. This allows motors to operate closer to their thermal limits without damage (reducing peak temperatures by 18%), potentially enabling short-duration over-torque maneuvers for noise abatement. Additionally, energy management strategies for hybrid architectures, such as those optimized by Artificial Gorilla Troop algorithms, can reduce fuel consumption and associated noise by 3–4%.

III. Analysis and Discussion

A. Methodology Comparison

A tension exists between experimental and computational methodologies. The NFAC wind tunnel tests provided high-fidelity acoustic data but suffered from wall reflections that required complex Vold-Kalman filtering to isolate propeller noise [3]. Conversely, CFD methods like those used by Liu et al. avoid reflection issues and provide full flow field visualization but often rely on turbulence models that may underpredict broadband noise [6]. A hybrid approach, calibrating lower-order aeroacoustic models with sparse experimental data, is necessary for rapid design iteration.

The NFAC tunnel is a 40-by-80-foot wind tunnel that has been used to test the acoustics of many different rotors. From these NFAC acoustic experiments, there has been a large amount of data on rotors and how they may differ in their acoustic signatures. These results are important to consider when discussing noise reduction of distributed electric propulsion because they give insights into potential new rotor designs that can both maximize performance and efficiency while also reducing noise allowing for operation in urban environments and at lower altitudes.

Computational fluid dynamics (CFD) is also important to consider in predicting the acoustic signatures of different rotors. For the NFAC experiments, the CFD simulations were carried out with the High-Performance Computer Modernization Program (HPCMP) using a rotorcraft simulation tool with the acoustics simulated by the noise prediction software PSU-WOPWOP [22].

From the NFAC tunnel testing, valuable information about acoustic differences between rotors’ acoustic signatures using experimental and CFD techniques was obtained. The plot in Fig. 4 comes from an experiment in the NFAC tunnel of the prop and wing configuration with the plot focusing on the isolated propeller which is ideal since the goal is to discern sound levels for rotors alone. The OASPL (overall sound pressure level, dB) for microphones 6–13 is shown for both Experiment (data obtained experimentally in the NFAC tunnel) and Iso Prop (CFD results).

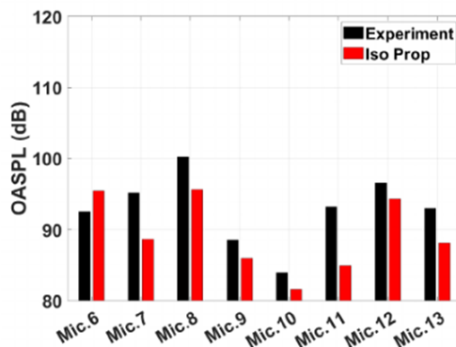


Fig. 4 Comparison of OASPL between the experiment and isolated propeller.

As seen in Fig. 4, there is an overprediction of sound levels for most cases which is likely due to sound wave reflections inside the NFAC tunnel that CFD does not have to deal with since those results are simulated in an environment

only consisting of the isolated propeller with no sources that could amplify acoustic results. These results also allow for comparison to other rotors, and using the NFAC tunnel to gather information about as many rotor types as possible is beneficial in understanding what rotors are best for reducing noise in UAVs, drones, etc.

There are some issues with relying on experimental results as perfectly gathered data. The poor panel absorption characteristics of the NFAC tunnel for rotors at low frequencies (below 100 Hz) may result in inaccuracies when obtaining acoustics data due to strong reflections that introduce distortions. Furthermore, low absorptency of the panels in the test section enclosure can lead to strong standing wave patterns that spatially amplify the amplitude of noise measurements at discrete frequencies associated with the geometry of the wind tunnel [23]. These issues can be seen in Fig. 5 showing test microphones 13 and 14 and how wall reflections may skew results slightly.

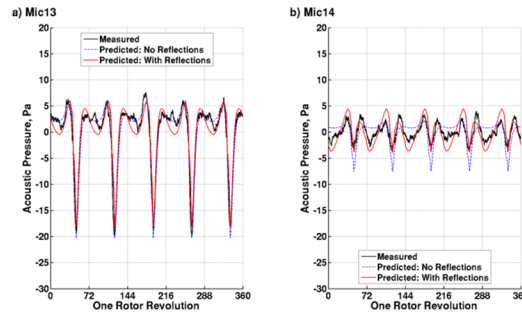


Fig. 5 Predicted time histories with and without reflections for microphones 13 and 14

CFD results are much faster to produce than doing so experimentally when using methods such as unstructured mesh CFD to model rotors. However, they may be less accurate because of these meshes and other factors. First, any small inaccuracies in geometry may create compounding effects regarding acoustics leading to quieter or louder noise results than expected. Second, while software is based on our physical understanding of aeronautic acoustics and tries its best to imitate real-world conditions, it can only go so far in simulating them because of how computationally intensive it is to have close-to-perfect simulations. So, even when using the most expensive and accurate programs, it is still very unlikely that the produced results will be close to the expected results.

Small Unmanned Aerial System (sUAS) acoustics have also been researched experimentally by NASA. The recorded data discussed occurred in the Cleveland National Forest, about 35 miles northeast of San Diego, in December of 2016. Flyover noise was recorded for an sUAS weighing 3.2 kg (unloaded) with level flyovers at 5 and 10 m/s and at 20, 30, 50, and 100 m above ground level (AGL). It is important to note that there were calm winds and little ambient noise affecting the results, which is important in minimizing errors in the experimental results. The experimental data gathered was compared to predictions of propeller or rotor noise. Two test campaigns were performed: the first in an anechoic chamber at the NASA Langley Research Center to assess the accuracy of pre-existing noise prediction techniques, and the second to study noise generating mechanisms of more realistic motor-vehicle configurations [24].

From the experiments described above, plenty of data was gathered pertaining to acoustic levels for many different types of rotor configurations. One such experiment provided experimental acoustic spectra in the form of sound pressure levels (SPLs) for multiple different separation distances between the rotor and the support arm (vertical separation). The rotor had a rotation rate of 5400 RPM with a blade passage frequency (BPF) of 180 Hz. The noise was measured using a single microphone below the rotor. Shown in Fig. 6 is a plot of SPL vs. the nondimensionalized parameter for separation distance between the rotor and the support arm.

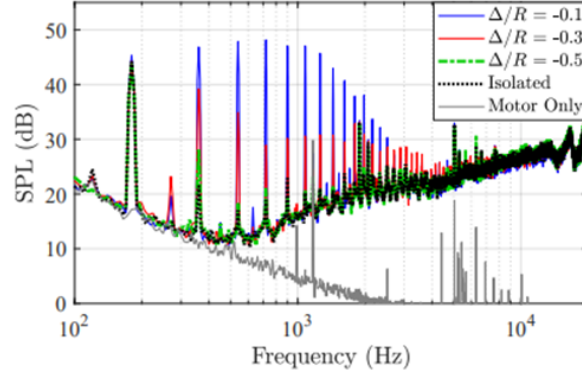


Fig. 6 SPL versus rotor-arm vertical separation

Fig. 6 shows a few trends. First, it can be observed that reducing the vertical separation between the rotor and the support arm dramatically increases SPLs of higher harmonics of the BPF. The case of least separation shows large increases in acoustic energy between BPFs while the case of highest separation only shows similarly large increases in acoustic energy at the 2nd and 3rd BPFs (360 and 540 Hz). Second, for the isolated rotor (no airframe), noise generated is dominated by a tone at the BPF with lower levels at higher harmonics of the BPF. These results show the impact of the airframe and its proximity to the rotor plane on sUAS acoustics. Higher vertical separation appears to be beneficial in reducing noise levels, so design of future sUAS that intend to operate with lower acoustic signatures should consider this characteristic.

One possible source of error for these results and limitation of the experiment may be only having one microphone which may result in less certain data since there are no other points to compare with the data gathered. It is also important to include multiple microphones to compare acoustic data for different distances from the rotor. Another limitation of this study is the lack of multiple rotor/airframe configurations used since more data on the differences between different rotors and airframes could be massively beneficial in understanding what design can reduce acoustic levels based on pre-existing rotors and/or airframe structures and if they can be implemented to decrease noise levels of currently existing sUAS and drones.

For a full quadcopter airframe, experimental results were gathered in both hover and forward flight conditions. These tests occurred at the NASA Langley Low Speed Aeroacoustic Wind Tunnel (LSAWT) which is an open-circuit free jet wind tunnel. The motor rotation rates and approximate relative rotor positions were monitored during the experiment using laser sensor tachometers. The two orientations tested were upright orientation and flyover orientation. The SPL vs. frequency plots for both configurations are shown in Fig. 7a and Fig. 7b.

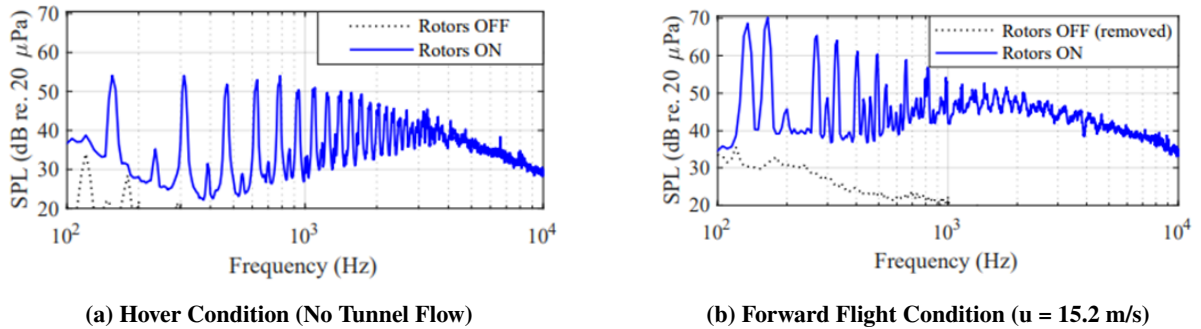


Fig. 7 Sound Pressure Level (SPL) Comparison for Different Flight Conditions.

As shown from the plots above, the forward flight condition produced much higher SPL values than the hover condition (about 15 dB higher in amplitude). Also, the frequency of SPL peaks for the forward flight condition is greater than that of the hover condition. These peaks are BPFs of the forward and aft rotation rates and are spaced evenly to keep thrust stable. Another important difference between these plots is that the hover condition acoustic levels do not fall off until at higher frequencies than the forward flight condition acoustic levels.

The trends described above may be explained by several factors. First, the hover condition requires less total thrust since all rotors are pointed downward compared to the forward flight condition which must produce enough thrust to propel itself at 15.2 m/s as well as maintain level flight requiring a larger thrust output from the motor. As a result, the rotor must operate at higher rates of rotation for the second case meaning higher acoustic levels are to be expected due to the higher rates of rotation. Second, the quicker fall off of the forward flight condition when compared to the hover condition for increasing frequency may occur due to the rotors already operating at a higher rotation, so the actual frequencies of rotors may be similar when drop off does occur for both cases when considering the differences in rotor speed.

This experiment has several limitations. First, like the experiment described beforehand, the lack of different rotors used limits the results as more rotor types could give much better insight into how rotor design impacts acoustic performance for both hover and forward flight conditions. Second, relying on vehicle-mounted laser sensor tachometers for measurements of the motor rotation rates and rotor locations may result in error due to vibrations caused by the motor and rotor in the airframe that could perturb the sensors.

Along with these experimental findings, computational methods were performed to predict hover noise. These methods include a high fidelity computational fluid dynamics (CFD) coupled with a Ffowcs Williams-Hawkings (FW-H) solver, a low-fidelity blade element analysis method for predicting tonal acoustics, and a semi-analytical frequency domain tool for predicting broadband rotor self-noise. The results produced dealt with isolated rotor, rotor-airframe interaction, and full vehicle noise predictions. These predictions are important to assess the accuracy of existing noise prediction techniques for rotors and rotor speeds typical of sUAS. This can be done by comparing computational results to experimental results.

For the isolated rotor noise predictions, high-fidelity CFD predictions were used to generate periodic impermeable surface pressure loading data that were input into a FW-H solver. The CFD code used was OVERFLOW2, an unsteady Reynolds-averaged Navier-Stokes (URANS) solver, and the FW-H solver utilized the PSU-WOPWOP code. The noise prediction technique can be referred to as OF2-PSW. The low-fidelity tonal acoustic predictions used the Propeller Analytic System (PAS) part of the NASA Aircraft Noise Prediction Program. The Broadband Acoustic Rotor Codes suite was used for the broadband noise predictions. Plots of predicted BPF are shown in Fig. 8a and Fig. 8b.

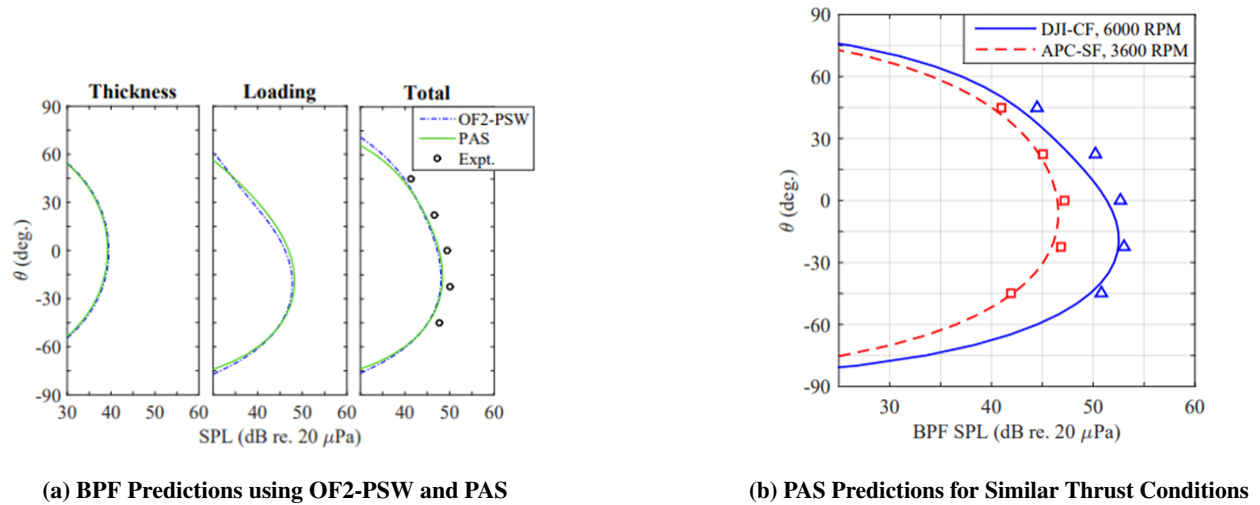


Fig. 8 Comparison of BPF Predictions by Observation Angle (θ).

Fig. 8a demonstrates that the OF2-PSW and PAS prediction techniques align well for both thickness and loading noise contributions. The total predicted noise predictions compare well with experimental data. Fig. 8b shows how PAS predictions vary for two different rotors of similar thrust conditions with considerably different geometries. These results help prove that PAS results are suitable for tonal acoustic characterization of isolated sUAS rotors in static hover conditions [24].

For the rotor-airframe interaction noise predictions, the OF2-PSW prediction technique was applied to the rotor-horizontal arm configuration. These results are compared to the experimental results in Fig. 9a. Furthermore, these noise contributions are broken up into the rotor and airframe in Fig. 9b. For the plots below, the case of the least vertical

separation is used ($\Delta/R=-0.1$) with a rotor rotation rate of 5400 RPM and microphone placement the same as that used for the experimental results of this setup.

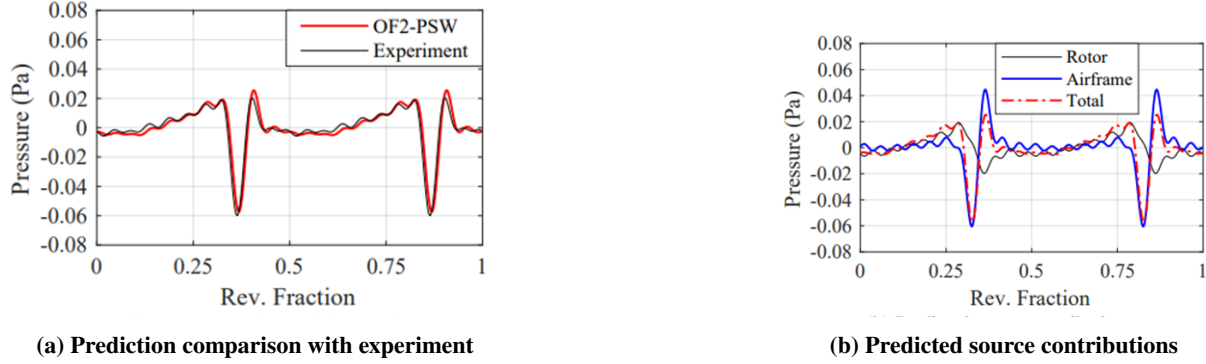


Fig. 9 Single-Rotor Acoustic Prediction Results.

From the above plots, it can be observed that the OF2-PSW predicted results compare extremely well to experimental results for the case of least vertical separation. This is important in showing the accuracy of CFD in predicting acoustic levels of sUAS and can be extremely useful in future design of rotors and airframes requiring low acoustic signatures as CFD with very close results to experimentally proven results can be obtained before ever testing them in the field. Based on the accuracy of the predicted results using OF2-PSW compared to the experimental results, the second plot can be used to predict relatively accurate experimental results for the noise contributions by the rotor and airframe as individual components. While the plot shows pressure vs. rev. fraction, the pressure values can be used to solve for SPL.

For the full vehicle noise predictions, the OF2-PSW prediction technique was once again used due to its accuracy to experimental results. A DJI Phantom-like vehicle was simulated with details shown in Fig. 10a. For this simulation, two cases are used: the first with rotors located a common distance above the airframe, and the second with rotors located a common distance below the airframe. The rotor rotation rate used is 5400 RPM, and the sUAS (rotor and airframe) is under hover conditions for both simulation cases. Acoustic predictions for an observer 45 degrees below the plane of rotor are shown in Fig. 10b. For the SPL vs. BPF Harmonic plot, the red diamonds represent the case of downward pointing rotor, and the blue squares represent the case of upward pointing rotor.

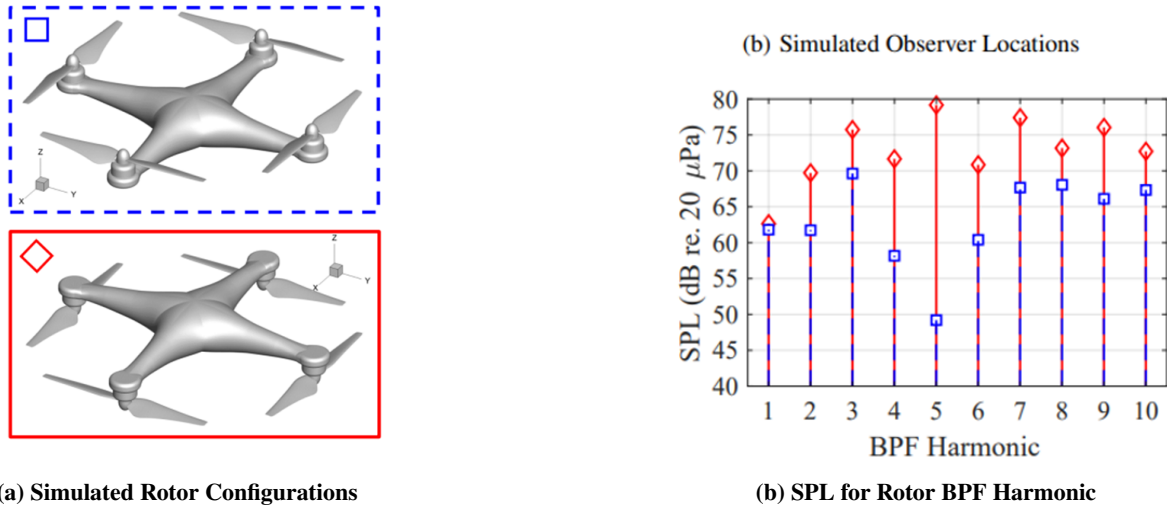


Fig. 10 Full Vehicle Acoustic Simulation Results. (a) Rotor placement cases analyzed. (b) Sound pressure level predictions at 45° below the rotor plane.

From the results in Fig. 10b, it can be observed that the case of rotors located below the airframe (pointed downward)

results in higher SPLs for all BPF harmonics of the rotor compared to the case of rotors located above the airframe (pointed upward). Also important to note is that for three observations, the OASPL defined as the summation of energy at the first ten BPF harmonics, shows that on average, the under-mounted motors are 8.1 dB louder (in OASPL) [24]. These results are important to note because they show how rotor placement (either above or below the airframe) contributes to different SPLs produced from the sUAS. This information is beneficial in design of sUAS because rotors pointed upward can be prioritized to limit acoustic signatures.

A possible limitation of these predicted results is their implementation in real-world scenarios due to lack of variance in wind, temperature, and other environmental factors in the predictions that may make the results less valuable when operating in conditions not explicitly used for the predictions. Another possible limitation, specifically for the rotor-airframe interaction noise predictions, is that only one case of the three for vertical separation is compared to experimental results. Predicting results for the other two cases would be more beneficial in comparing the two to see how the predicted results hold up to experimental for differing vertical separations. Having closeness in predicted and experimental results for multiple different vertical separations would further back the accuracy of the predicted results.

B. BTMS Architecture Trade-offs in DEP Context

The viability of DEP for UAM hinges upon the effective thermal management of high-energy-density lithium-ion battery packs. Unlike ground-based electric vehicles, eVTOL aircraft experience “peaky” load profiles characterized by extreme power demands during Vertical Take-Off (VTO) and Vertical Landing (VL) phases, followed by moderate cruise loads [2]. This transient thermal behavior introduces unique challenges that cannot be addressed by simply scaling automotive battery thermal management solutions. Recent research has established that eVTOL battery discharge rates during hover operations can reach 2C to 5C, generating heat fluxes an order of magnitude greater than typical electric vehicle applications [15]. The fundamental heat generation within lithium-ion cells is governed by the electrochemical processes occurring during discharge, expressed as:

$$\dot{Q}_{gen} = I(t)^2 R_{int}(T, SOC) + T \Delta S \frac{I(t)}{nF} \quad (12)$$

where $I(t)$ is the instantaneous current, R_{int} is the internal resistance dependent on temperature T and State of Charge (SOC), ΔS represents the entropy change, n is the number of electrons transferred, and F is Faraday’s constant. The first term represents irreversible Joule heating, while the second captures reversible entropic contributions. For eVTOL operations, the high current demands during takeoff maximize the $I^2 R$ losses, creating rapid thermal transients that the BTMS must manage without excessive mass penalty.

Li et al. provide a comprehensive taxonomy of thermal management architectures for eVTOL applications, categorizing systems based on their energy consumption and heat transfer mechanisms [2]. Passive systems rely on natural convection, radiation, or phase change without external power input. These include air cooling with extended fin surfaces, Phase Change Materials (PCM), and heat pipes. Active systems utilize external power to force convection or enable refrigeration, encompassing forced air cooling, pumped liquid loops, vapor compression refrigeration, and thermoelectric cooling. Hybrid systems combine passive and active mechanisms to leverage the high specific energy storage of passive components with the controllability of active systems.

The selection of an appropriate BTMS architecture for DEP applications must account for the unique operational constraints of vertical flight. Unlike ground vehicles where weight penalties translate primarily to reduced efficiency, in aircraft the mass of the cooling system directly impacts the thrust requirements and, consequently, the acoustic signature. This coupling creates a design optimization challenge where thermal performance cannot be evaluated in isolation from aerodynamic and acoustic considerations. The following chart highlights the trade-offs and the different architectures along with some performance metrics.

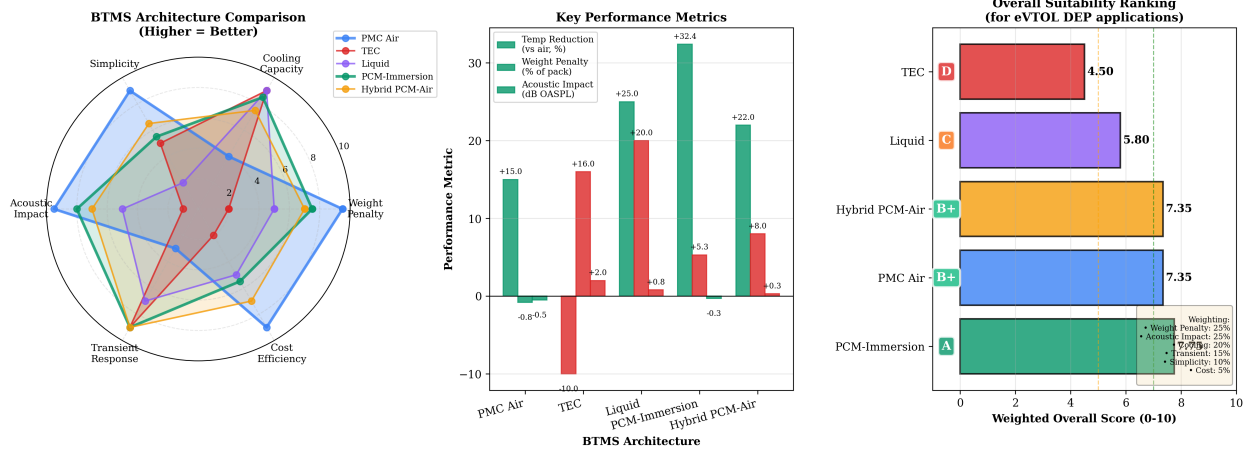


Fig. 11 Comprehensive comparison of battery thermal management system architectures for eVTOL DEP applications. (Left) Multi-dimensional performance across six metrics. (Middle) Quantitative performance data from literature. (Right) Weighted overall suitability ranking. PCM-Immersion hybrid systems emerge as optimal for balancing thermal performance with acoustic constraints.

1. Lightweight Passive Architectures: Advanced Materials and Geometries

The pursuit of lightweight BTMS solutions has driven research into advanced materials and geometric innovations. Nelson and Almansour at NASA Glenn Research Center investigated PMC battery packs with helical fins as an alternative to traditional aluminum heat sinks [11]. The PMC approach offers several advantages for aerospace applications. These composites utilize a polymer base doped with thermally conductive fillers such as pitch-based carbon fibers or boron nitride ceramics. Unlike isotropic metals, PMCs can be engineered to exhibit anisotropic thermal conductivity, directing heat preferentially along the fin direction while maintaining electrical insulation in other directions.

The NASA study achieved a weight reduction of 0.04 kg per module compared to conventional aluminum designs, translating to an 8.25 Wh/kg increase in effective pack energy density. For a large DEP vehicle carrying 500 kg of batteries, this efficiency gain is substantial, potentially enabling either extended range or reduced thrust requirements with corresponding acoustic benefits. The geometric innovation of helical fins addresses the fundamental limitation of straight-fin designs where thermal boundary layer growth progressively degrades heat transfer efficiency downstream. The helical geometry induces a swirling flow component that continuously disrupts boundary layer development, maintaining high convective heat transfer coefficients throughout the fin length. This results in significantly enhanced Nusselt numbers compared to straight-fin configurations.

However, Nelson et al. concluded that PMC air cooling has inherent limitations for high-power eVTOL applications. The relatively low specific heat capacity and thermal conductivity of air constrain the maximum achievable heat rejection rate. For vehicles requiring high-power takeoffs or operating in elevated ambient temperatures, air cooling, even with optimized PMC, the heat sinks may be insufficient to maintain battery temperatures below safety thresholds during the 90–120 second takeoff transient.

2. Active Thermoelectric Cooling: Precision at a Cost

Thermoelectric Coolers (TECs) based on the Peltier effect represent an alternative approach that offers precise temperature control without moving fluids. Kellermann et al. investigated the viability of TEC-based BTMS for a 19-seat hybrid-electric aircraft [16]. The thermoelectric approach offers advantages including solid-state reliability, immunity to the gyroscopic and vibrational loads inherent to tilting DEP rotor mechanisms, and the capability to both heat and cool the battery pack depending on ambient conditions.

However, the Kellermann study revealed severe weight penalties associated with thermoelectric cooling. Their optimization analysis indicated that implementing a full TEC-based BTMS results in a Maximum Take-Off Weight (MTOW) penalty of 2% to 16%. This penalty arises not from the TEC modules themselves but from cascading system-level effects. TECs typically exhibit Coefficients of Performance between 0.5 and 1.0 for high temperature differentials, meaning that removing 1 kW of waste heat requires consuming approximately 1 kW of electrical power.

The downstream heat sink must then dissipate both the original battery heat and the electrical work input to the TEC, effectively doubling the heat rejection requirement. Furthermore, the electrical consumption represents a parasitic load that drains the main battery, necessitating larger battery capacity to achieve the same range, which in turn generates more heat.

The acoustic implications of a 16% MTOW increase are substantial. Referring to the fundamental equation of lift in hover, such a mass increase requires a commensurate increase in thrust. To achieve this higher thrust without modifying rotor geometry, the flight controller must increase rotor RPM, driving up tip Mach numbers and causing a disproportionate increase in acoustic intensity. This finding suggests that thermoelectric cooling may be limited to auxiliary or localized sub-cooling roles rather than serving as the primary BTMS architecture for noise-sensitive urban missions.

More recent work by Zhang et al. on thermoelectric cooling systems specifically for eVTOL batteries operating on hot days demonstrated that TEC-based BTMS can enable operation at ambient temperatures up to 45°C [17]. However, this capability comes at the expense of reduced flight cycles and additional weight, reinforcing the fundamental trade-off between thermal performance and vehicle mass.

3. Hybrid PCM-Immersion Cooling: Peak Shaving for Transient Loads

Recognizing the limitations of both pure passive and pure active approaches, Chai et al. proposed a hybrid architecture specifically tailored to the “peaky” DEP load profile, combining Phase Change Materials (PCM) with dielectric fluid immersion cooling [4]. This system exploits the complementary characteristics of each technology across different flight phases.

During takeoff, when the battery generates maximum heat, the immersion fluid rapidly transfers thermal energy to the PCM. The PCM melts, absorbing this energy as latent heat of fusion while maintaining a near-constant temperature at the phase transition point. Critically, this heat is stored rather than immediately rejected to the ambient environment. During cruise and after landing, when power demands are reduced, the cooling system slowly rejects the stored heat to atmosphere via a compact radiator, re-solidifying the PCM for the subsequent flight cycle. After landing, immersion cooling restores the solid state of the PCM, preparing the system for the next mission.

Chai et al. validated this system experimentally, demonstrating a 32.4% reduction in maximum cell temperature compared to baseline air cooling. Numerical simulations on a small-scale eVTOL battery module showed that the PCM weight accounted for only 5.32% of the total battery pack weight while maintaining battery pack temperatures below 48°C with an initial temperature of 25°C across multiple flight cycles. The maximum temperature difference within the battery cells during discharge was held to 4.74°C, satisfying the uniformity requirements for lithium-ion pack operation.

The “peak shaving” mechanism offers a weight advantage over conventional liquid cooling systems. Without PCM, the radiator and pump must be sized to handle the maximum takeoff heat load—potentially 50 kW or more for large eVTOL vehicles. With PCM absorption, the radiator can be sized for the much lower average mission load, perhaps 5–10 kW. Although the immersion fluid adds mass, the reduction in heat exchanger size and associated hardware can result in a net weight benefit compared to fully-sized active liquid loops.

The Chai architecture also addresses the distributed nature of DEP propulsion systems. Rather than requiring long fluid lines from a central fuselage reservoir to wingtip-mounted battery packs, self-contained PCM/immersion modules can be located at each nacelle, eliminating pumping losses and leak risks associated with distributed liquid cooling networks. However, this approach increases the rotational inertia of wing-mounted structures, potentially affecting aeroelastic flutter characteristics and the actuation power required for tilt-wing mechanisms.

4. Liquid Cooling Systems: Performance and Weight Trade-offs

Channeled liquid cooling technology remains a prominent approach for high-power eVTOL applications. Recent work on wavy-channel liquid cooling systems has demonstrated that liquid-based BTMS can maintain battery operating temperatures within acceptable limits with a system mass of less than 20% of the battery pack weight [14]. The system architecture involves circulating coolant through wavy channels attached to battery cells, with heat transported to a plate-fin compact heat exchanger for dissipation to atmosphere.

Sensitivity analyses of the heat exchanger model reveal the impact of design parameters on BTMS size, weight, and parasitic power consumption. Optimization of the coupled wavy-channel and heat exchanger system demonstrates that liquid cooling can reduce battery degradation by over three times compared to air-cooled systems for both tilt-wing and lift-plus-cruise eVTOL configurations. However, the weight of the cooling system, particularly the coolant mass and heat exchanger, remains a significant consideration for acoustic-sensitive applications.

Chen et al. investigated the optimization of liquid-cooled BTMS for a notional tilt-wing reference eVTOL aircraft, examining the effects of cruise altitude, hover duration, and battery pack oversizing on BTMS weight and battery lifetime [15]. Their results revealed that varying cruise altitude has a negligible effect on BTMS design, primarily because the dominant thermal challenge occurs during the high-power hover phases near ground level. However, extended hover duration significantly impacts BTMS weight requirements, necessitating considerably heavier cooling systems to manage accumulated thermal energy. Battery oversizing emerged as a strategy to mitigate capacity degradation, but at the cost of increased vehicle mass with corresponding acoustic penalties.

5. Heat Pipe Integration for Passive Heat Acquisition

Flat heat pipes integrated with ram air cooling represent an emerging approach for eVTOL thermal management that leverages the high effective thermal conductivity of two-phase heat transfer without the weight and complexity of pumped liquid systems. Wang et al. proposed a BTMS combining flat heat pipes with ram air for a short-range eVTOL aircraft, analyzing the system through numerical simulations [13].

The flat heat pipe configuration offers advantages for prismatic battery cell geometries common in eVTOL packs. Heat is conducted from the battery cells to the evaporator section of the heat pipe, where the working fluid vaporizes and travels to the condenser section exposed to ram airflow during flight. The phase change process enables effective heat transport with minimal temperature differential along the heat pipe length.

Experimental studies on the thermal characteristics of 61.5 Ah lithium-ion batteries under flight discharge conditions informed the design of this passive cooling system. The effects of ram air inlet flow rate, temperature, and fin spacing on thermal management performance were systematically evaluated. Results indicated that the heat pipe-based system can effectively manage thermal loads during cruise phases when ram air is available, but may require supplementary cooling during stationary ground operations when natural convection is the only available mechanism.

6. Optimization Frameworks and Algorithmic Design

The complexity of BTMS design for DEP applications—involving thermal, structural, mass, and acoustic constraints—has motivated the development of advanced optimization frameworks. Wiston et al. employed a hybrid optimization strategy combining Genetic Algorithms (GA) with Simultaneous Perturbation Stochastic Approximation (SPSA) to navigate the multi-objective BTMS design space [19].

The optimization problem formulation encompassed multiple objectives: minimizing system mass (M_{sys}), minimizing parasitic energy consumption (E_{para}), and minimizing peak temperature (T_{max}). Constraints included maintaining cell temperature below 60°C and pack temperature uniformity within 5°C. Design variables encompassed coolant type, flow rates, heat sink geometry, PCM mass fraction, and pump power.

The two-stage optimization methodology leverages the complementary strengths of each algorithm. The GA performs a bio-inspired global search, exploring the entire disjoint design space including discrete architectural choices between air, liquid, and PCM-based systems. Once promising regions are identified, SPSA refines the continuous variables within those architectures. SPSA is particularly effective for thermal optimization because thermal simulations are computationally expensive and inherently noisy; calculating exact gradients for every variable would be prohibitive. SPSA approximates gradients by perturbing all variables simultaneously, enabling rapid convergence.

Through this GA+SPSA methodology, Wiston et al. identified optimized configurations achieving a 22% reduction in peak battery temperature for a given weight budget compared to baseline heuristic designs. This thermal margin can be exploited either for enhanced safety factors or to enable high-power noise-abating maneuvers that would otherwise be thermally constrained.

Guibert et al. extended optimization to the structural domain, proposing thermo-mechanical level-set topology optimization for eVTOL battery packs [20]. This multi-physics approach simultaneously considers the thermal loads from battery heat generation and the mechanical loads arising from battery pack placement within the vehicle structure. At the system level, a linear elastic beam model captures the boom-mounted configuration typical of many eVTOL designs. At the battery scale, physics-based electrochemical models predict heat generation based on mission power requirements. The optimization objective seeks a thermally and structurally efficient lightweight structure, recognizing that battery pack mass directly influences both thermal loads (through required capacity) and mechanical loads (through structural weight).

C. The Thermal-Acoustic Design Paradox

The synthesis of thermal management literature reveals a fundamental “Thermal-Acoustic Paradox” that constrains DEP vehicle design. This paradox emerges from the opposing requirements of acoustic and thermal optimization, mediated through the vehicle mass budget. The thermal-acoustic paradox (Figure 12) demonstrates why conventional BTMS approaches fail. As shown in Figure 11, hybrid PCM-immersion systems offer.

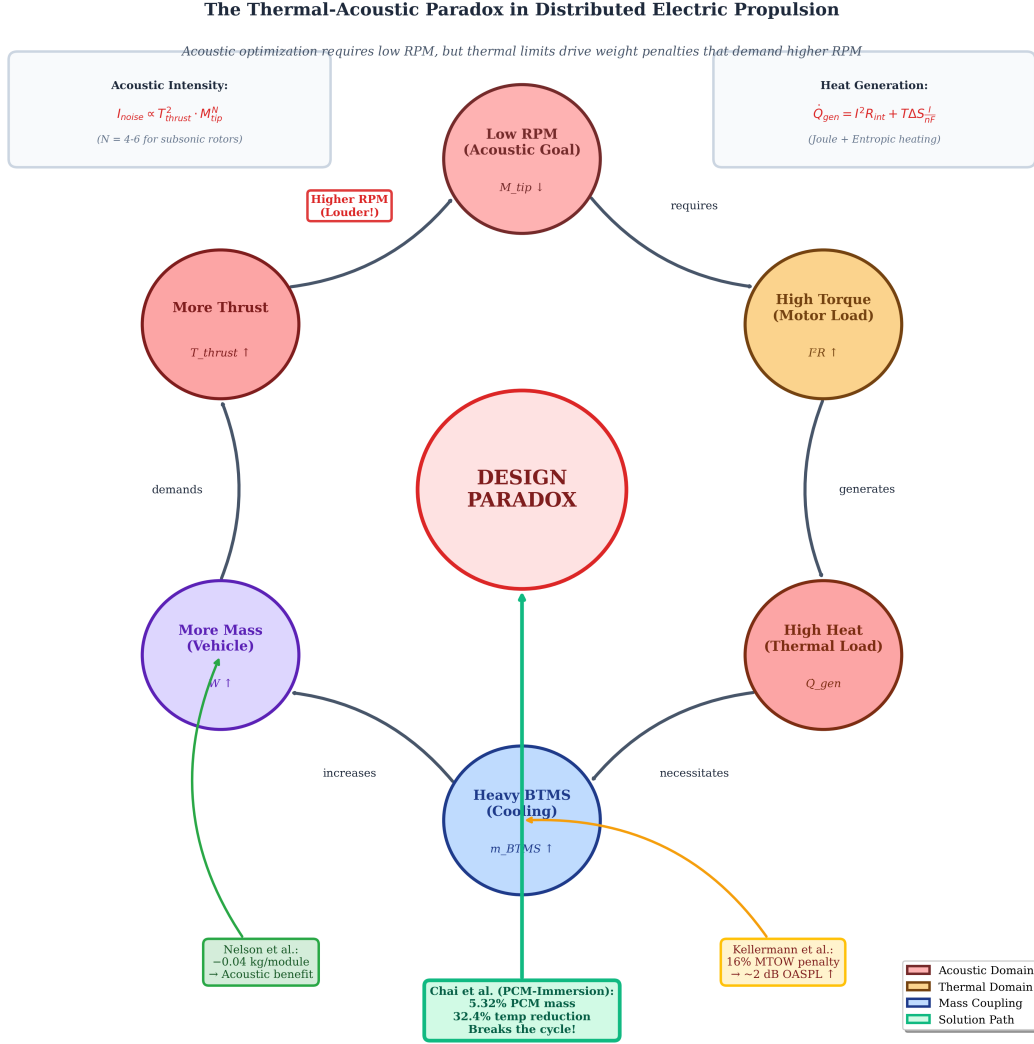


Fig. 12 The Thermal-Acoustic Paradox in distributed electric propulsion. Acoustic optimization favors low-RPM operation to minimize tip Mach number, but the resulting high motor torque generates excessive heat. Heavy cooling systems add mass, forcing higher thrust requirements that demand increased RPM—precisely undermining the acoustic goal. Annotations show quantitative impacts from recent studies.

1. The Physics of Coupled Design

Acoustic optimization for community acceptance generally favors large, slow-turning rotors to minimize tonal noise and tip vortex interaction. Low tip-speed operation reduces the tip Mach number (M_{tip}), which strongly influences the acoustic intensity scaling:

$$I_{noise} \propto T_{thrust}^2 M_{tip}^N \quad (13)$$

where N typically ranges from 4 to 6 for subsonic rotors. However, generating the thrust required for vertical flight at low RPM necessitates high torque output from the electric motors. The resistive heating in motor windings scales with

current squared (I^2R losses), and current is proportional to torque for a given motor [25]. Thus, low-RPM operation maximizes the thermal challenge.

Addressing this thermal load requires either passive systems with sufficient thermal mass (heavy) or active systems with adequate heat rejection capacity (also heavy, plus parasitic power consumption). The added BTMS mass increases the aircraft’s gross take-off weight:

$$W_{aircraft} = W_{structure} + W_{payload} + W_{batteries} + m_{BTMS} \quad (14)$$

An increase in m_{BTMS} requires an increase in hover thrust. For fixed rotor geometry, achieving higher thrust requires increasing rotor RPM, which drives up M_{tip} and causes a disproportionate increase in acoustic intensity. The paradox is thus closed: thermal mitigation adds mass, mass requires more thrust, and more thrust generates more noise—precisely the outcome that acoustic optimization sought to avoid.

2. Quantifying the Trade-off

The reviewed literature enables quantification of this trade-off. Kellermann et al. demonstrated that a thermoelectric BTMS can impose up to 16% MTOW penalty [16]. Using the fundamental hover equation and assuming constant disk loading, this translates to a 16% increase in required thrust. For typical rotor noise scaling ($N \approx 5$), and assuming RPM increases proportionally to thrust for fixed-geometry rotors, the acoustic intensity could increase by a factor of $(1.16)^2 \times (1.16)^{5/2} \approx 1.6$, or approximately 2 dB OASPL.

Conversely, the lightweight PMC approach of Nelson et al. saves 0.04 kg per module [11]. For a 500 kg battery pack comprising approximately 100 modules, this represents a 4 kg reduction in BTMS mass—roughly 0.4% of a 1000 kg eVTOL vehicle. While seemingly modest, such savings are cumulative across all thermal management components and represent the correct design philosophy for acoustic-sensitive applications.

The hybrid PCM-immersion approach of Chai et al. offers a middle path [4]. The PCM mass penalty (5.32% of battery pack weight) is offset by reduced heat exchanger requirements, potentially achieving net weight parity with air-cooled systems while providing superior thermal performance. This “peak shaving” strategy explicitly acknowledges the transient nature of eVTOL thermal loads and designs the BTMS to store rather than immediately reject the takeoff heat spike.

3. Toward Multi-Disciplinary Design Optimization

Current research often optimizes thermal, aerodynamic, and acoustic disciplines in isolation. The literature clearly indicates that this siloed approach is insufficient for DEP vehicles where these domains are tightly coupled. Future research must focus on Multi-Disciplinary Design Optimization (MDO) frameworks that treat BTMS mass as a direct variable in the acoustic objective function.

An integrated MDO framework would link the BTMS mass directly to the acoustic solver through the following coupled loop: (1) Define the mission profile and thermal environment; (2) Calculate battery heat generation and select BTMS architecture, determining m_{BTMS} ; (3) Calculate required hover thrust including BTMS mass; (4) Evaluate rotor noise based on thrust and tip speed requirements; (5) If noise exceeds community acceptance limits, iterate by either increasing rotor size (aerodynamic change), switching BTMS architecture (thermal change), or altering flight path (trajectory change).

Bisht et al. have begun developing such frameworks, employing thermo-electrochemical modeling to predict mission-specific BTMS demands and their implications for battery degradation and thermal safety [26]. Their work identifies critical phase-dependent demands on lithium-ion batteries, demonstrating that landing and balked-landing phases can trigger thermal safety limits through increased current densities. Cognizant of these mission-specific interactions, practical insights into the dynamic response of battery thermal management systems can inform next-generation vehicle design.

The development of intelligent control strategies further extends the design space. Shili et al. developed a “Novel Intelligent Thermal Feedback Framework” using machine learning to estimate motor winding temperatures in real-time [21]. This approach allows motors to operate closer to their thermal limits without damage, reducing peak temperatures by 18% and potentially enabling short-duration over-torque maneuvers for noise abatement. By exploiting the transient thermal capacity of propulsion components, such intelligent control laws can expand the low-noise flight envelope without requiring additional BTMS mass.

IV. Conclusion

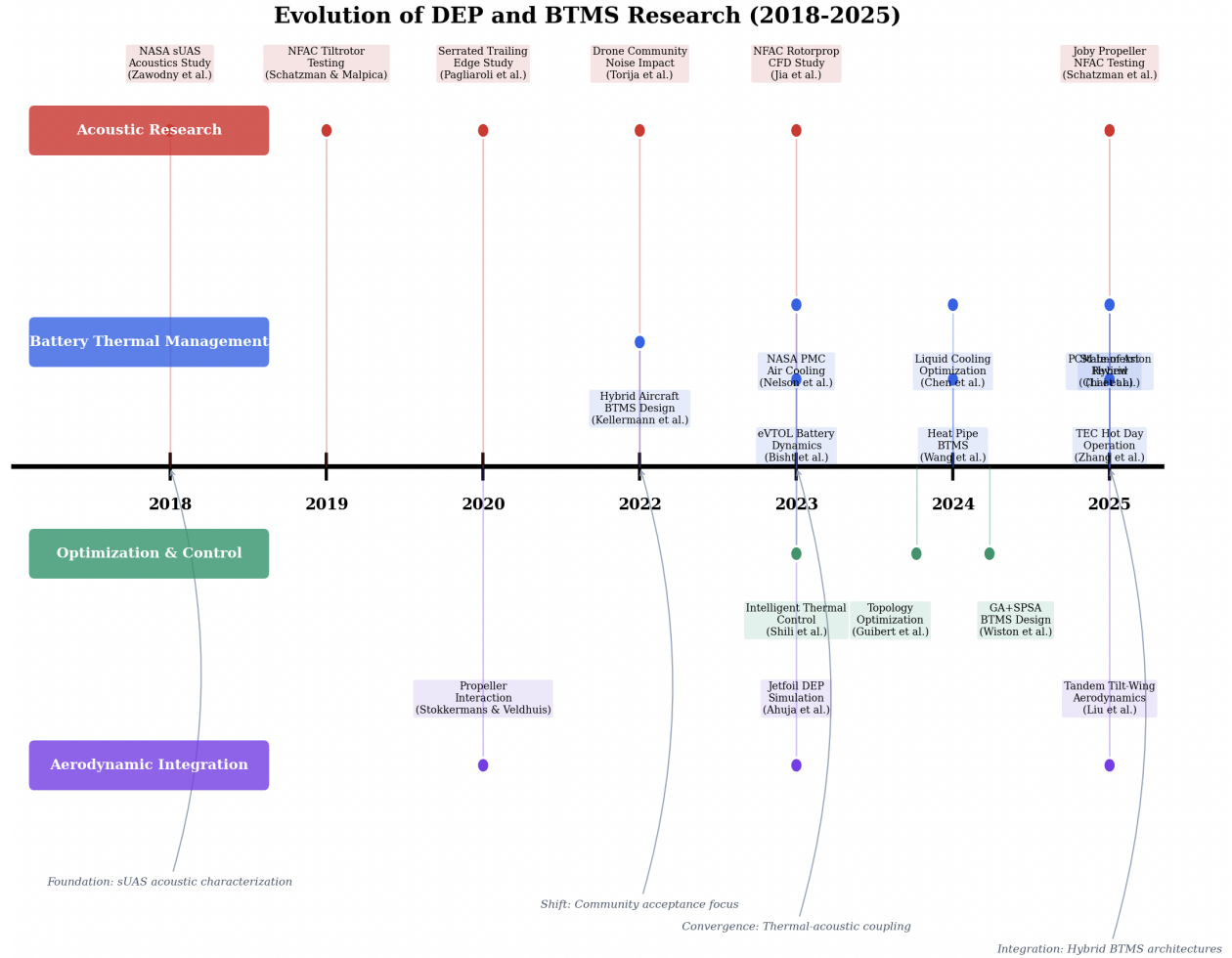


Fig. 13 Research Timeline showcasing advances made in DEP systems

Figure 13 shows the progress made in this field over the past couple of years. Reducing noise in DEP systems requires a holistic approach that extends beyond aerodynamics to include thermal and control subsystem design. While aero-propulsive coupling and novel configurations like ducted fans offer pathways to lower noise, they are constrained by the thermal limits of current battery and motor technology. The weight penalties associated with effective thermal management systems directly counteract acoustic benefits by necessitating higher thrust. Future research must focus on coupled aero-thermal-acoustic optimization and the development of intelligent control laws that exploit the transient thermal capacity of propulsion components to expand the low-noise flight envelope.

References

- [1] Tyto Robotics, “Distributed Electric Propulsion: A Comprehensive Guide,” Tech. rep., Tyto Robotics, 2025. <https://doi.org/10.2514/6.1986-1802>.
- [2] Li, Z., Xie, P., Lin, C., and Liu, G., “A state-of-the-art review on eVTOL thermal management: system architectures, key components and emerging technologies,” *eTransportation*, Vol. 26, 2025. <https://doi.org/10.1016/j.etrans.2025.100480>.
- [3] Schatzman, N. L., Weist, L., Bain, J. J., and Thai, A. D., “Acoustic Testing of the Joby Aviation Propeller in the National Full-Scale Aerodynamics Complex 40- by 80-Foot Wind Tunnel,” , 2025. <https://doi.org/10.2514/6.2025-0844>.
- [4] Chai, X., Xu, X., and Liang, X., “Strategy of battery thermal management for eVTOL by combination of PCM and immersion cooling,” *Thermal Science and Engineering Progress*, Vol. 64, 2025. <https://doi.org/10.1016/j.tsep.2025.103824>.
- [5] Carlos Ramos-Romero Nathan Green Seth Roberts Charlotte Clark Antonio J. Torija, “Requirements for Drone Operations to Minimise Community Noise Impact,” Tech. rep., International Journal of Environmental Research and Public Health, 2022.
- [6] Liu, J., Li, G., Ma, D., and Wang, F., “Aerodynamic interference and wake characteristics of a multirotor tandem tilt-wing electric vertical take-off and landing aircraft in conversion flight,” *Physics of Fluids*, Vol. 37, No. 11, 2025. <https://doi.org/10.1063/5.0302547>.
- [7] Stokkermans, T., and Veldhuis, L., “Aerodynamic Interaction Effects Between Propellers in Typical eVTOL Vehicle Configurations,” Tech. rep., Delft University of Technology, 2020.
- [8] Ahuja, V., Lakshminarayan, V., and Alvarez, E., “Simulations of Distributed Electric Propulsion Concepts and Integration with a Novel Airframe,” Oak Ridge Leadership Computing Facility User Meeting, 2023. Whisper Aero and ORNL.
- [9] Pagliaroli, P. C. R. E. N. F. P. T., “Experimental Analysis of Small-Scale Rotors with Serrated Trailing Edge for Quiet Drone Propulsion,” *Journal of Physics: Conference Series*, 2020.
- [10] Gur, O., “Multidisciplinary Design Optimization of a Quiet Propeller,” *ResearchGate*, 2008. URL https://www.researchgate.net/publication/268583758_Multidisciplinary_Design_Optimization_of_a_Quiet_Propeller.
- [11] Nelson, W., Almansour, A. S., and Singh, M., “Design and Analysis of Battery Thermal Management Systems,” , 2023. NASA Glenn Research Center.
- [12] Santhosh, S., Kumar, A., and Patel, V., “Advances in battery thermal management for electric vehicles: A comprehensive review of hybrid PCM-metal foam and immersion cooling technologies,” *Renewable and Sustainable Energy Reviews*, Vol. 208, 2025, p. 115021. <https://doi.org/10.1016/j.rser.2024.115021>.
- [13] Wang, J., Li, G., and Zhang, F., “Research on battery heat generation characteristics and thermal management system applied to a typical eVTOL,” *Applied Thermal Engineering*, Vol. 255, 2024, p. 123856. <https://doi.org/10.1016/j.applthermaleng.2024.123856>.
- [14] Xiong, R., Ye, Z., and Wang, C., “Liquid Cooling Systems for Batteries of Electric Vertical Takeoff and Landing Aircraft,” *Journal of Aircraft*, Vol. 61, No. 3, 2024, pp. 892–905. <https://doi.org/10.2514/1.C037404>.
- [15] Chen, X., Zhang, H., and Wang, Y., “Optimisation of a liquid cooling system for eVTOL aircraft: Impact of sizing mission and battery size,” *Applied Thermal Engineering*, Vol. 247, 2024, p. 123054. <https://doi.org/10.1016/j.applthermaleng.2024.123054>.
- [16] Kellermann, H., Fuhrmann, S., Shamiyeh, M., and Hornung, M., “Design of a Battery Cooling System for Hybrid Electric Aircraft,” *Journal of Propulsion and Power*, Vol. 38, No. 5, 2022, pp. 726–740. <https://doi.org/10.2514/1.B38695>.
- [17] Zhang, T., Chen, Q., Xu, M., Zhang, D., and Fei, Q., “Thermoelectric cooling systems for eVTOL batteries operating on hot days,” *Applied Thermal Engineering*, Vol. 263, 2025, p. 125903. <https://doi.org/10.1016/j.applthermaleng.2025.125903>.
- [18] Dong, W., Shen, J., Xu, Y., Wu, Z., Xu, X., Long, Y., Wang, X., and Zhou, F., “Performance Analysis of Hybrid Thermal Management System Integrating Air-Cooling Topology Fin and Phase Change Material for eVTOL Aircraft Battery,” *Energy Conversion and Management*, 2025. <https://doi.org/10.2139/ssrn.5358384>.
- [19] Wiston, P., Chen, L., and Kumar, R., “Battery Thermal Management Design Space Exploration for eVTOL Applications,” *Applied Thermal Engineering*, Vol. 252, 2024. <https://doi.org/10.1016/j.applthermaleng.2024.123456>.
- [20] Guibert, A. T. R., Bookwala, M., Cronk, A., Meng, Y. S., and Kim, H. A., “Thermo-Mechanical Level-Set Topology Optimization of an eVTOL Battery Pack,” *AIAA SciTech Forum*, 2024. <https://doi.org/10.2514/6.2024-2362>.

- [21] Shili, M., Hammedi, S., Chaoui, H., and Nouri, K., “A Novel Intelligent Thermal Feedback Framework for Electric Motor Protection in Embedded Robotic Systems,” *Electronics*, Vol. 14, No. 18, 2025. <https://doi.org/10.3390/electronics14183598>.
- [22] Zhongqi (Henry) Jia, R. J., Joon W. Lim, “Computational Study on the Aerodynamic and Acoustic Rotorprop Test in the NFAC 40- by 80-Foot Wind Tunnel,” Tech. rep., US Army CCDC AvMC, 2023. <https://doi.org/10.4050/F-0079-2023-17945>.
- [23] Sim, N. L. B. C. K. B. W., “Acoustics Reflections of Full-Scale Rotor Noise Measurements in NFAC 40- by 80-Foot Wind Tunnel,” Tech. rep., Nasa, 2013. URL <https://ntrs.nasa.gov/api/citations/20130011016/downloads/20130011016.pdf>.
- [24] Nikolas S. Zawodny, R. C., Andrew Christian, “A Summary of NASA Research Exploring the Acoustics of Small Unmanned Aerial Systems,” Tech. rep., NASA, 2018. URL <https://ntrs.nasa.gov/api/citations/20180002208/downloads/20180002208.pdf>.
- [25] Solo Motor Controllers, “Thermal Effect on Brushless Motors,” Tech. rep., Solo Motor Controllers, 2025. URL <https://www.solomotorcontrollers.com/blog/thermal-brushless-motors/>.
- [26] Bisht, A., Amin, R., Dixit, M., Wood, N., Kweon, C. B. M., and Belharouak, I., “Revealing hidden predicaments to lithium-ion battery dynamics for electric vertical take-off and landing aircraft,” *Joule*, Vol. 7, No. 9, 2023, pp. 2016–2034. <https://doi.org/10.1016/j.joule.2023.07.014>.